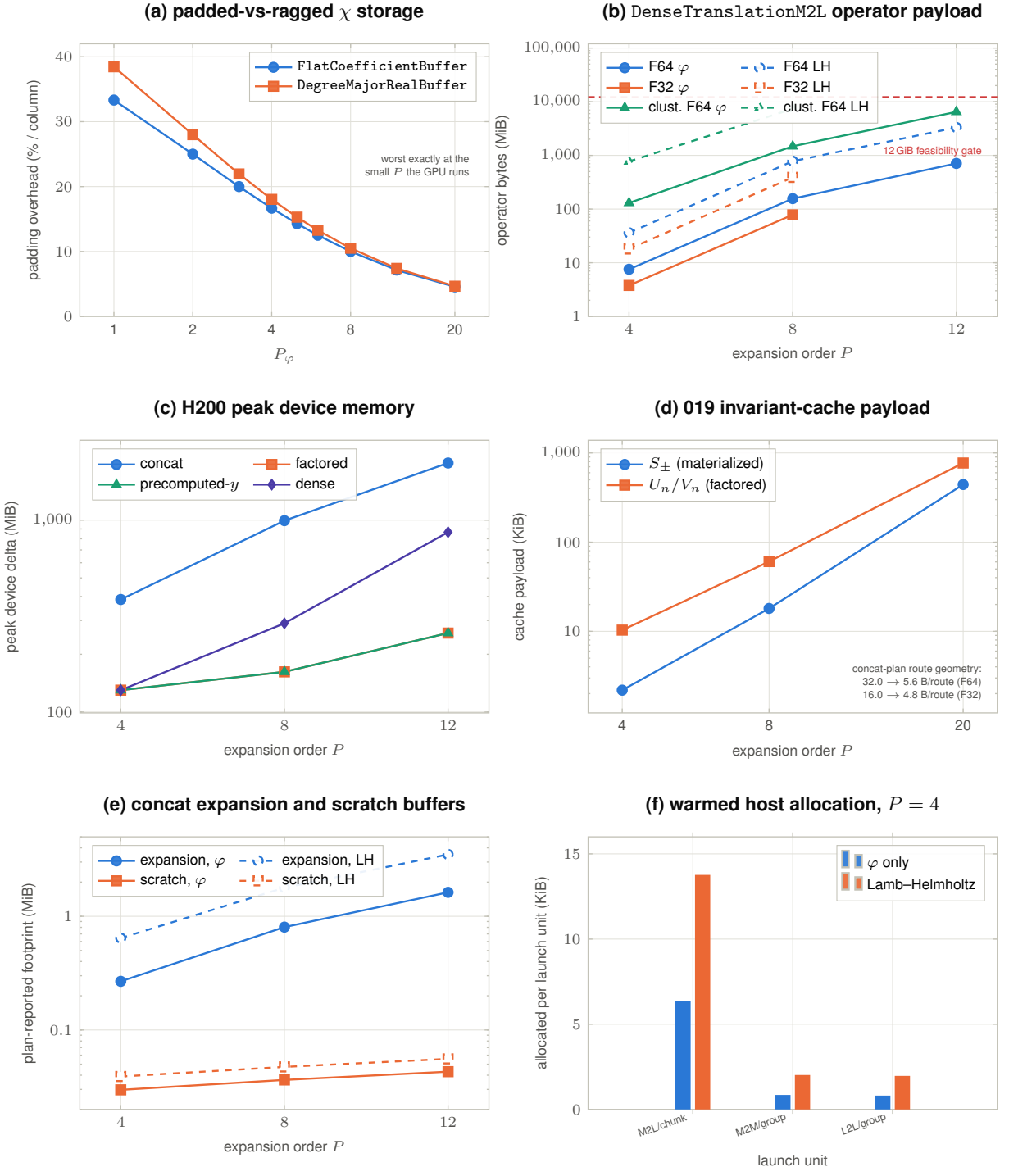
Fig. 4 — Storage and allocation: buffers, operators, caches, geometry, and warmed launches



Machines / regimes: (a) analytic, host-independent. (b,c) `m13h-1-1` H200, task 024 CUDA campaign, uniform $n = 20000$ except the clustered series. (d,f) local macOS measurement, task 019. Panel (e) uses the task-024 H200 concat plan at uniform $n = 20000$, Float64. **Data:** `smallp_fallback_layout/m12-1-7/layout_storage.csv`, `operator_ab_benchmark/summary/memory_feasibility.csv`, `operator_performance_tuning/local_macos_allocations_storage.csv` (tables `fig04a_*` …`fig04g_*`). **Reads as:** padding χ to P_φ costs 17–33% (flat) / 18–39% (degree-major) extra storage per column at $P \leq 4$ — worst precisely where the GPU operates — which is why the ragged layout was kept. The dense strategy's operator payload grows by roughly $20\times$ per doubling of P and crosses the 12 GiB gate for Lamb–Helmholtz at $P = 12$ under clustering (30.1 GiB) — the 21 infeasible cases of task 024. Panel (c) is the counter-intuitive part: for φ-only $n = 20000$ the *highest* measured device peak is the whole-slab concat path ($405 \rightarrow 2082$ MiB), not dense ($136 \rightarrow 908$ MiB), because concat's peak is dominated by slab scratch rather than by operators, while factored and precomputed-y stay flat at 136–271 MiB. Dense becomes the memory-limited strategy only once P, Lamb–Helmholtz or clustering inflate its operator payload past the gate, as in panel (b). Panel (e) separates the concat plan's coefficient storage from its reported operator scratch: expansion storage grows from 0.27 to 1.62 MiB for φ and from 0.64 to 3.50 MiB for Lamb–Helmholtz over $P = 4 \rightarrow 12$, while this plan's reported scratch remains 0.03–0.06 MiB. Panel (f) records the remaining warmed host dynamic overhead after tuning: 6.4 KiB per M2L chunk and 0.8–0.9 KiB per M2M/L2L group for φ, rising to 13.8 and about 2.0 KiB with χ. **Caveats:** panel (c) uses `peak_bytes`, which on CUDA is a per-strategy device-memory delta and therefore comparable; the CPU `peak_bytes` column is a process-wide `Sys.maxrss()` shared by all four strategies and is deliberately not plotted. Likewise `persistent_bytes`/`scratch_bytes` are plan-reported for dense and precomputed-y but fall back to `Base.summarysize` for concat and factored, so panel (e) is deliberately a within-concat order/channel view, not a cross-strategy scratch comparison. Panel (f)'s unit is stage-specific (chunk for M2L, group for M2M/L2L), as recorded by the source campaign; it compares warmed launch overhead, not total stage allocation. Only dense reports a non-zero `operator_bytes`: the other three build no materialized per-class operator. In panel (c) the factored and precomputed-y curves coincide exactly at all three orders (136/170/271 MiB) — the device peak is quantised by the CUDA allocation pool at this scale, so the orange squares sit underneath the green triangles. Dense Float32 at $P = 12$ is a hard-coded skip in the 024 harness, not an observed failure, so that series ends at $P = 8$.